

Rishab Dey

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www.linkedin.com/in/rishab-dey | [GitHub](#) | [Notion](#) | [Kaggle](#) | [Portfolio : rishabdey.github.io](#)

SUMMARY

Data Scientist (M.Sc.) with 3+ years of experience as a Data Analyst, applying machine learning and statistical methods across research and industry settings. Skilled in predictive modeling, time-series analysis, and explainable AI. Experienced in building automated data pipelines and interactive dashboards to support data-driven decision-making. Proficient in Python, SQL, and TensorFlow, with expertise in model evaluation, feature engineering, and interpretability.

SKILLS

Programming: Python, R, SQL (MySQL, PostgreSQL)

Libraries: NumPy, Pandas, Scikit-learn, TensorFlow, Statsmodels, Matplotlib, Seaborn, Plotly, SpaCy

Machine Learning & Statistics: Time Series Analysis, Statistical Modeling, Feature Engineering, Model Optimization, Model Evaluation, Explainable AI, Generative AI (LLMs, Prompt Engineering, RAG, LangChain, Transformers)

Visualization & BI: Streamlit, Power BI, Tableau, Looker Studio, Folium, Geospatial Analysis

Data Systems & MLOps: MLflow, dbt, Git, DataPipeline Development, Data Transformation, DataQuality Management

Other: Web scraping (Selenium, BeautifulSoup, rvest), MATLAB, Adobe Photoshop, Figma, PLC Programming, Arduino

WORK EXPERIENCE

Data Analyst Student Assistant, Fraunhofer-Institut für Materialfluss und Logistik (Sustainability and Circular Economy)

Feb 2023 – March 2026
Dortmund, Germany

- Designed and deployed a secure multi-user analytics dashboard to replace manual Excel-based sustainability reporting, automating GHG and KPI calculations for 120 companies with over 1,500 multi-year data submissions.
- Processed and analyzed international trade datasets on plastics and rubber flows to and from Germany, building interactive geospatial visualizations to identify material transfer patterns and cross-border logistics trends.
- Designed and deployed a real-time environmental monitoring system integrating IoT sensor streams (Arduino) with a Python-based analytics and visualization layer, enabling live tracking of temperature, location, speed, and UV metrics to support Digital Product Passport (DPP) initiatives and enhance lifecycle transparency.

Tools: Python, Streamlit, Pandas, NumPy, Plotly, Folium, Seaborn, Matplotlib, Arduino MKR WIFI 1010, MKR Environmental shield

Data Analyst Intern, Breezeitech Services Pvt. Ltd.

Feb 2019 – March 2020
Mumbai, India

- Analyzed inventory and product performance data for networking and surveillance products (routers, CCTV systems) to evaluate cost efficiency, defect rates, and sales trends, supporting pricing and procurement decisions.
- Performed exploratory and statistical analysis on monthly spending and product performance data, developing dashboards and reports to identify trends and opportunities for process improvement.

Tools: Python, NumPy, Pandas, Plotly, Matplotlib, Seaborn.

EDUCATION

Master of Science in Data Science, [Universität Potsdam](#) 

Oct 2021 – Feb 2026
Potsdam, Germany

Master Thesis: Explainable AI for Time Series Monitoring of Cocoyam Drying Using Hyperspectral Imaging

Key coursework: Machine Learning, Statistical Learning for Big Data, Advanced Statistical Learning, Computer Vision, Optimization, Linear Models, Learning in Robotics, Inference

PROJECTS

Source-Aware Multi-Agent Research Pipeline → [↗](#)

Developed a multi-agent system that generates structured research reports from a user query with explicit source attribution. Orchestrates agents for web search via Tavily API, content extraction (BeautifulSoup), report generation, and evaluation. Implemented a Streamlit interface with real-time pipeline tracking and downloadable outputs.

Tools: Python, LangChain, Mistral AI, Tavily API, BeautifulSoup, Streamlit

Explainable AI for Time Series Monitoring Using Hyperspectral Imaging (Master's Thesis)

Developed a CNN-LSTM model to predict moisture content from hyperspectral near-infrared time-series data, achieving an R^2 of 0.87. Applied Grad-CAM and FusionGrad for model interpretability and feature attribution, enabling improved feature selection through the identification of key spectral bands and spatial regions.

Tools: Python, TensorFlow, NumPy, Pandas, Scikit-learn, Hyperspectral Imaging, Grad-CAM, FusionGrad, HPC (4× NVIDIA Tesla V100 GPUs)

Image Classification using Deep Learning & Dimensionality Reduction

Built a CNN-based image classifier on the Fashion-MNIST dataset and compared performance across multiple optimizers. Applied PCA, Kernel PCA, and t-SNE for dimensionality reduction and cluster visualization to analyze feature separability.

Tools used: Python, TensorFlow, Keras, Scikit-learn, NumPy, Pandas, Matplotlib, Seaborn, Kaggle

Player Statistics (Top 5 European Football Leagues) → [↗](#)

Built and deployed an interactive football analytics dashboard using Streamlit by scraping player performance data from [FBref.com](#) [↗](#) across Europe's top five leagues. Designed a PostgreSQL database to store and manage 33,000+ records covering over 3,000 players and enabled player and league comparisons across key performance metrics through interactive visualizations. Tools: Python, PostgreSQL, Pandas, NumPy, Plotly, Streamlit, Selenium

WorldLytics : Interactive Data Insights → [↗](#)

Developing an evolving analytics portfolio featuring interactive visualizations and comparative insights across real-world datasets to explore trends and support data-driven storytelling. Tools: Python, Pandas, Plotly, Figma, Adobe Photoshop, Notion.

CERTIFICATES

- [Introduction to MLflow \(DataCamp\)](#) [↗](#)
- [Intermediate dbt \(DataCamp\)](#) [↗](#)
- [Introduction to dbt \(DataCamp\)](#) [↗](#)

LANGUAGES

- English (C1)
- German (A2, actively improving towards B1/B2 for professional use)

PAPERS AND PRESENTATIONS

Explainable AI-Driven Sparse Hyperspectral Modeling for Moisture Forecasting During Cocoyam Drying

Manuscript submitted for publication (2026), (LWT) Lebensmittel-Wissenschaft & Technologie

Conference on Advances in Electronics Engineering (CAEE 2018),

Presented and published a paper on 'Waste Management System' - ISBN Number: 978-0-9994483-6-6

ACHIEVEMENTS

- Captain of Junior College Football Team,
- Taekwondo Black Belt 1st Dan , Sang Rok World, Recognized by World Taekwondo Federation
- Rapporteur, International Press and Research Team (Model United Nations, held at the University)
- First Prize in Second Year Bachelors Project